UNDERSTANDING GLOBAL HAPPINESS: BEYOND ECONOMIC GROWTH

CONTEXT

Traditional economic indicators such as GDP are often used to evaluate national progress. However, they fail to capture broader aspects of human wellbeing such as social support, trust in institutions, freedom, and health.
The United Nations World Happiness Report, based on survey data collected through the Gallup World Poll, provides a multidimensional measure of life satisfaction across countries.

Problem Statement
This study aims to:
- Identify clusters of countries with similar wellbeing characteristics
- Predict national happiness scores using socio-economic indicators

Research Questions
- Do countries form distinct wellbeing clusters based on social and economic factors?
- Which variables most strongly influence happiness levels across nations?

Significance
Understanding these relationships helps policymakers design strategies that improve societal wellbeing beyond economic growth.

DATASET OVERVIEW

Dataset Source

World Happiness Report (United Nations)

DATA CHARACTERISTICS AND KEY VARIABLES

1. Observations: ~150 countries
2. Time frame: recent global survey data
3. Data collected through international life satisfaction surveys

Variable	Description
Happiness Score	Overall life satisfaction index
GDP per Capita	Economic prosperity indicator
Social Support	Perceived availability of help from community
Healthy Life Expectancy	Health and longevity measure
Freedom	Ability to make life choices
Generosity	Measure of social altruism
Perception of Corruption	Trust in institutions

Why This Dataset Matters

The dataset provides a multidimensional view of wellbeing, enabling analysis of economic, social, and institutional factors affecting happiness.

METHODOLOGY (SUPERVISED LEARNING APPROACH)

To understand the factors influencing national happiness levels, a supervised learning approach was implemented.

Model Used: Random Forest
Random Forest is an ensemble machine learning algorithm that builds multiple decision trees and aggregates their predictions to improve accuracy and reduce overfitting.
Target Variable: Happiness Score

Input Features

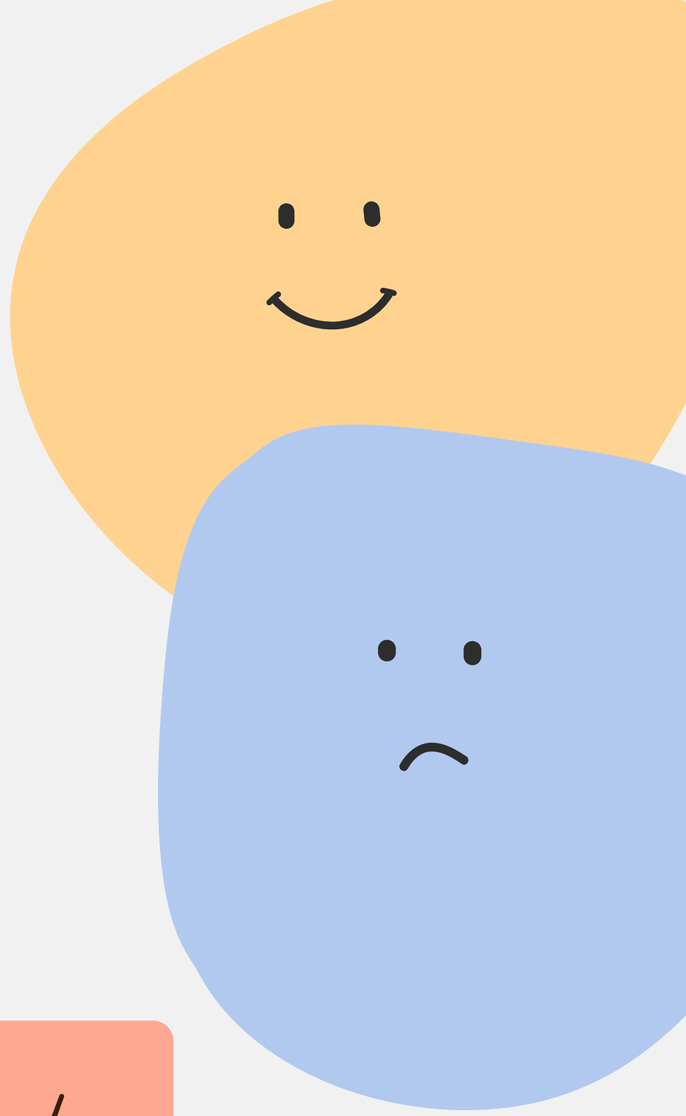
- GDP per capita
- Social support
- Healthy life expectancy
- Freedom to make life choices
- Generosity
- Perception of corruption

Model Workflow

1. Data preprocessing and normalization
2. Train–test split of dataset
3. Model training using Random Forest
4. Prediction of happiness scores
5. Evaluation using performance metrics

This supervised learning model allows us to identify which socio-economic factors most strongly influence national wellbeing.

HOW THE OUTPUT HELPS



The model indicates that the strongest predictors of happiness are:

1. Once the model is trained, it provides two important outputs.
2. First, it predicts happiness scores for countries based on the input variables.
3. Second, it shows feature importance, meaning which variables contribute most to predicting happiness.
4. From previous research using this dataset, factors like social support, freedom to make life choices, and healthy life expectancy tend to be strong predictors.
5. This helps us understand that happiness is influenced by social and institutional factors, not just income levels.
6. These insights can help identify key drivers of wellbeing across countries.

The model demonstrates that wellbeing is multidimensional, influenced by both economic prosperity and social infrastructure.

SOCIAL & ECONOMIC IMPLICATIONS

The findings highlight that national wellbeing depends on social, institutional, and health-related factors, not solely economic growth.

Policy Recommendation 1
Strengthening Social Support Systems
Investments in community programs, social safety nets, and mental health services can improve overall wellbeing.

Policy Recommendation 2
Promoting Institutional Transparency
Reducing corruption and strengthening governance increases trust and societal satisfaction.

Policy Recommendation 3
Improving Public Health Infrastructure
Policies focused on healthcare access and life expectancy directly contribute to higher happiness levels.
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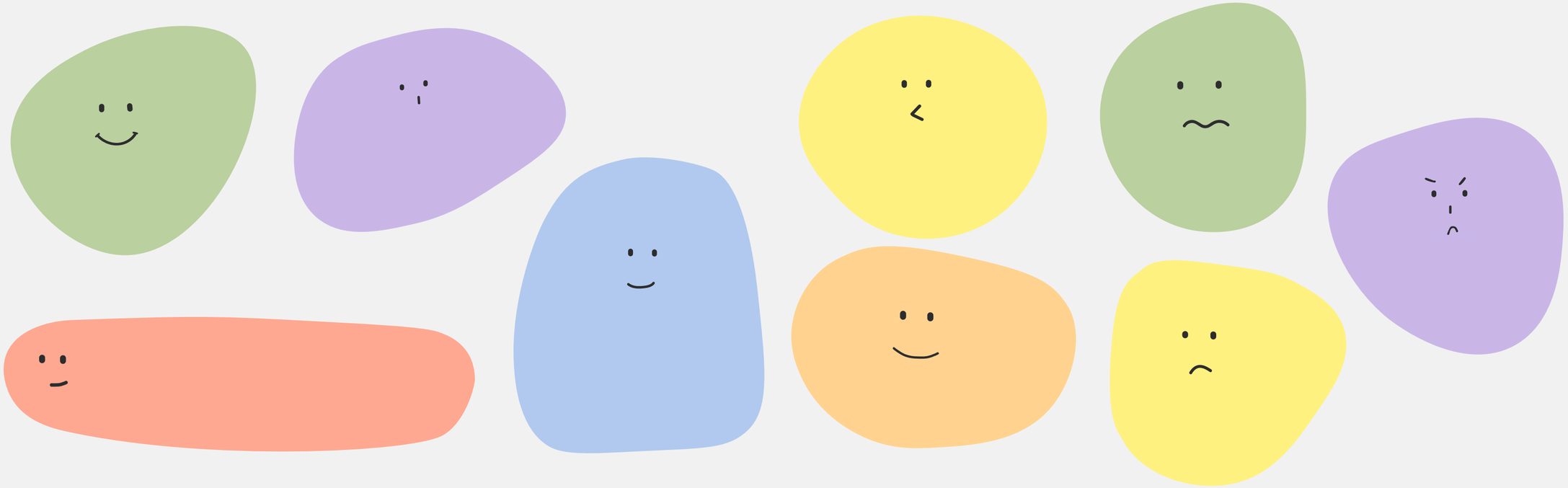
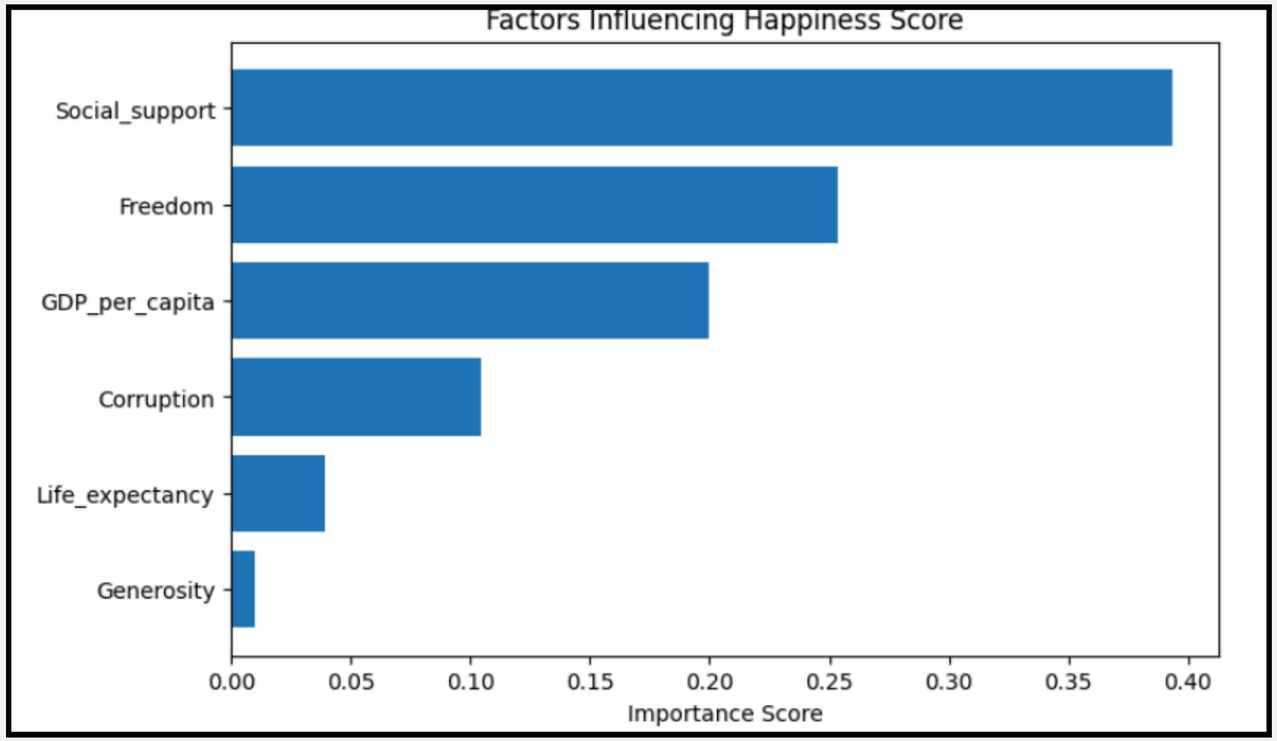
Section A

THANK YOU



OUTPUT FROM CODE

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   Generosity  Corruption  Happiness_score
0        0.15        0.03              7.8
1        0.18        0.04              7.6
2        0.20        0.05              7.5
3        0.16        0.06              7.4
4        0.19        0.05              7.3

Prediction Results:
    Actual Happiness Score  Predicted Happiness Score
27                     4.9                     4.7475
15                     7.2                     7.1345
23                     5.8                     5.6340
17                     6.4                     6.3570
8                      7.0                     7.0050
9                      6.8                     6.7590

Factor Importance Ranking:
            Factor  Importance Score
1   Social_support          0.393157
3          Freedom          0.253852
0   GDP_per_capita          0.200005
5       Corruption          0.104366
2  Life_expectancy          0.038947
4       Generosity          0.009674
```